

Volume III. Second Book. Quadratic Fields

Ninth Section. Discriminant

§ 84. Definition of discriminants

In the theory of quadratic fields there occur, as discriminants, certain rational integers, positive or negative, which, if they are even, are divisible by 4, and, if they are odd, leave the remainder 1 when divided by 4. Kronecker called such numbers “numbers of discriminant form.” We shall here call them simply discriminants, without being disturbed by the fact that the word has many other meanings in algebra. We define:

1. A positive or negative non-zero integer which satisfies one of the two congruences

$$D \equiv 0, \quad D \equiv 1 \pmod{4} \quad (1)$$

is called a discriminant.

Every square number is, by this definition, a discriminant. Since, however, squares occupy a certain exceptional position, we shall call them improper discriminants when a distinction is necessary.

2. The product of two, and consequently of any number of, discriminants is again a discriminant.

3. A discriminant which, apart from 1, contains no quadratic divisor after whose removal a discriminant remains is called a fundamental discriminant. Fundamental discriminants will subsequently be denoted by Δ . If D is not a fundamental discriminant, there is one and only one square Q^2 and one fundamental discriminant Δ such that

$$D = Q^2 \Delta. \quad (2)$$

Then Δ is called the fundamental discriminant, or the stem, of D .

A fundamental discriminant is divisible by no odd square. If, however, Δ is divisible by 4, then necessarily

$$\Delta \equiv 8, 12 \pmod{16}. \quad (3)$$

For if $\Delta \equiv 0, 4 \pmod{16}$, then after removing the factor 4 a discriminant would still remain. Thus, in order to find the stem of a given D , one first removes the greatest odd square divisor and then a sufficiently high power of 4, until the remaining Δ is either odd and $\equiv 1 \pmod{4}$, or is $\equiv 8$ or $12 \pmod{16}$.

4. A proper discriminant which is not decomposable into factors which themselves are discriminants is called a prime discriminant. If p is an odd natural prime number, and the sign is so chosen that $\pm p \equiv 1 \pmod{4}$, then the prime discriminants are

$$\pm p, -4, +8, -8. \quad (4)$$

5. Every fundamental discriminant can be decomposed in one and only one way into prime discriminants. For if one separates from Δ first all factors $\pm p$, only one of the numbers $1, -4, +8, -8$ remains.

§ 85. The extended Legendre-Jacobi symbol

The Legendre symbol $\left(\frac{m}{p}\right)$, when m is any positive or negative non-zero integer and p is a natural prime number not dividing m , has the value $+1$ if m is a quadratic residue of p , and the value -1 if m is a quadratic non-residue of p . Jacobi extended the meaning of the symbol so that a composite number can stand in place of p . If m is a discriminant, however, a still further extension is often convenient.

If D is a discriminant and n an integer, we define a symbol (D, n) by the following rules:

$$(D, 0) = 0, \quad (1)$$

$$(D, 1) = 1, \quad (2)$$

$$(D, -1) = \begin{cases} 1, & D > 0, \\ -1, & D < 0, \end{cases} \quad (3)$$

$$(D, p) = 0 \quad \text{if } p \mid D, \quad (4)$$

$$(D, 2) = (-1)^{(D^2-1)/8}, \quad D \text{ odd}, \quad (5)$$

$$(D, p) = \left(\frac{D}{p} \right) \quad (p \text{ odd prime, } p \nmid D). \quad (6)$$

If $\pm n$ is decomposed into prime factors,

$$\pm n = p_1 p_2 \cdots p_r,$$

then

$$(D, n) = (D, \pm 1)(D, p_1)(D, p_2) \cdots (D, p_r). \quad (7)$$

The symbol is thereby defined without contradiction for all n , and has only the values $-1, 0, +1$. It is zero precisely when D and n are not relatively prime. From (7) follows

$$(D, nn') = (D, n)(D, n'), \quad (8)$$

and, if q^2 is a square prime to n ,

$$(q^2 D, n) = (D, n). \quad (9)$$

The formulas belonging to the law of reciprocity take a special form in this notation. If $\pm p \equiv 1 \pmod{4}$, then for every n not divisible by p ,

$$(\pm p, n) = \left(\frac{\pm p}{n} \right). \quad (10)$$

For odd n one has likewise

$$(-4, n) = (-1)^{(n-1)/2}, \quad (8, n) = (-1)^{(n^2-1)/8}, \quad (-8, n) = (-1)^{(n-1)/2 + (n^2-1)/8}. \quad (11)$$

Further, if D, D' are two discriminants, then

$$(DD', n) = (D, n)(D', n). \quad (12)$$

The reciprocity law with its supplementary theorems can then be summarized as

$$(D, D') = \pm (D', D), \quad (13)$$

where the upper sign holds when at least one of the two discriminants D, D' is positive, and the lower sign when both are negative.

Since every square is a positive discriminant, one obtains, if m is prime to D ,

$$(D, m^2) = 1, \quad (14)$$

and, together with (8),

$$(D, mn^2) = (D, m) \quad (15)$$

whenever mn and D have no common divisor. Moreover

$$(D, n) = (D, n') \quad \text{if } n \equiv n' \pmod{D}. \quad (16)$$

If n, n' in (16) are themselves discriminants, one obtains from (13) and (16)

$$(D_1, D) = \pm(D_2, D), \quad (17)$$

where the lower sign occurs only when the two numbers D and $D_1 D_2$ are both negative. If D_1, D_2 are odd and $D = 4m$, this gives

$$(D_1, m) = \pm(D_2, m), \quad (18)$$

provided

$$D_1 \equiv D_2 \pmod{4m}. \quad (19)$$

The same formula holds in general. The formulas (16) and (18) can be used to compute the symbol (D, n) by the algorithm of the greatest common divisor.

If D is not a square, one can always determine a number β such that

$$(D, \beta) = -1. \quad (21)$$

For if Δ is the stem of D , and β is prime to D , then $(D, \beta) = (\Delta, \beta)$. Decomposing Δ into prime discriminants, one chooses β to make one factor give -1 and the others give $+1$. Therefore, if s runs through a complete system of residues modulo D , and if

$$S = \sum_s (D, s),$$

then multiplication by such a β gives also $S = -S$, and therefore

$$S = 0. \quad (22)$$

Thus in a complete system of incongruent numbers prime to D there are as many numbers α with $(D, \alpha) = +1$ as numbers β with $(D, \beta) = -1$.

§ 86. The Gaussian sums

In the first volume we became acquainted with the Gaussian sums from the theory of cyclotomy. These expressions can be generalized, and, by using the symbol (D, n) , take a simple form.

It was shown previously that, if n is an odd prime number, if s is not divisible by n , and if k runs through a complete residue system modulo n , then

$$\sum_k \left(\frac{k}{n}\right) e^{2\pi i s k / n} = \left(\frac{s}{n}\right) \sqrt{\pm n}, \quad (1)$$

with the usual choice of sign according as $n \equiv 1$ or $3 \pmod{4}$. Put $\Delta = \pm n \equiv 1 \pmod{4}$. Then Δ is a prime discriminant, and, by §85,

$$(\Delta, k) = \left(\frac{\Delta}{k}\right) = \left(\frac{k}{n}\right),$$

with the corresponding supplementary signs included. Formula (1) may therefore be written

$$\sum_k (\Delta, k) e^{2\pi i s k / \Delta} = (\Delta, s) \sqrt{\Delta}, \quad (2)$$

where $\sqrt{\Delta}$ is positive real or positive imaginary according as Δ is positive or negative. Direct calculation shows that the same formula also holds for

$$\Delta = -4, \quad +8, \quad -8,$$

and hence for every prime discriminant.

Assume the formula true for two relatively prime fundamental discriminants Δ' and Δ'' . Since

$$\sqrt{\Delta'}\sqrt{\Delta''} = \pm\sqrt{\Delta'\Delta''}, \quad (3)$$

where the lower sign occurs only when both factors are negative, put

$$k = k'\Delta'' - k''\Delta'.$$

As k', k'' run through complete residue systems modulo Δ', Δ'' , the number k runs through a complete residue system modulo $\Delta = \Delta'\Delta''$. Multiplication of the two sums gives

$$\sum_k (\Delta, k) e^{2\pi i sk/\Delta}. \quad (5)$$

Using §85, (16) and the reciprocity formula §85, (13), one obtains

$$(\Delta, k) = (\Delta', k')(\Delta'', k''),$$

and consequently (2) is valid for the product $\Delta'\Delta''$. By decomposition into prime discriminants, formula (2) is therefore valid for every fundamental discriminant Δ .